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GRES: Guaranteed Remaining Energy Scheduling of Energy-harvesting Sensors by Quality Adaptation

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***Abstract*—Many battery-powered IoT sensor nodes rely on harvesting energy which must be assumed an unreliable source. Previous works have shown that a sensor node can adapt its power consumption to keep its battery's state of charge at a sufficient level to achieve perpetual operation by, e.g., dynamically adapting its duty cycle. In this paper, we show that it is also possible to reduce the energy consumed by the operation of a sensor node by controlling the quality of the processed and transmitted data. Moreover, whereas most state-of-the-art methods rely on forecasting energy harvesting, risking loss of service in case of wrong predictions, this paper presents an algorithm called Guaranteed Remaining Energy Scheduling (GRES) which dynamically controls the quality of processed and transmitted data of a sensor node at runtime based on the state of charge of the battery, and providing a guarantee of safe continuous operation at the expense of data quality despite fluctuations in expected harvested energy. In experiments, GRES is evaluated and compared to an approach computing ILP-generated quality schedules for one full day ahead based on the assumption of a perfect harvested energy prediction. It is shown that the latter approach is not only computationally and energy-wise expensive, but also can lead to power shutdowns in case of wrongly predicted harvesting profiles.**

***Index Terms*—IoT, Power Management, Energy Harvesting**

I. Introduction

Many IoT applications run on battery-powered wireless sensor nodes, some of which can operate for years on coin cell batteries without replacement. Only some of these sensors have a known workload and volume of data produced per day, such that based on an estimation of their energy consumption, their lifetime can be predicted, i.e., the duration before a battery switch becomes necessary.

However, with the advent of edge computing, more compute-intensive tasks (e.g., neural network inferences) are being offloaded to sensor nodes, along with necessary higher energy requirements. To increase (possibly indefinitely) their lifetime, many sensor nodes therefore sustain their battery by harvesting energy from an environmental source (e.g., photovoltaic panels, kinetic energy). These energy sources, however, are not always reliable and their availability may fluctuate heavily, even within a day.

Therefore, it is necessary to adapt the running configuration of a sensor, e.g., power modes, at runtime to compensate for these fluctuations. Research on energy harvesting sensor nodes focuses on developing policies that aim at Energy Neutral Operation (ENO), i.e., within a given time window, adapting the energy consumed to the energy produced in order to leave the sensor in the same state of charge as it was at the start. This can be achieved, e.g., by generating a schedule of the configurations beforehand (as in [1]), reactively, e.g., deciding on the next configuration (as in [2], [3]) based on observed energy harvested, or a combination of both (as in [4]). The authors of [1] while laying the foundations of ENO, present a method to adapt the duty cycle (i.e., the amount of time spent sleeping during a periodic execution) to satisfy this constraint while maximizing a defined Quality of Service (QoS), e.g., the average latency of data transmission.

Many of these approaches like [5] require an accurate prediction of the harvested energy until a certain time horizon, e.g., a day, and have the goal of maximizing the quality of service during this time by balancing the energy consumed by the sensor node. However, forecasting the harvested energy can incur computational costs (when performed locally) or communication costs (when offloaded, e.g., to a cloud). Or, additional sensors (e.g., a weather station) are necessary to fine-tune the said prediction. In the worst case, mispredictions of a harvested energy profile may lead to abrupt power failures causing temporary data loss. Additionally, in the case of solar energy harvesting, today's challenges as posed by climate change lead to more and more risks of mispredictions [6].

In this paper, we present Guaranteed Remaining Energy Scheduling (GRES), a novel algorithm for dynamically controlling the quality of processed and transmitted data of a sensor node at runtime based on the current battery level and, different from other proposed approaches, providing a guarantee of safe continuous operation over a given time horizon at the expense of data quality despite fluctuations in expected harvested energy. Eventually, we compare the results obtained by our algorithm with optimal schedules obtained by an ILP for assumptions of non-accurately predicted traces of harvested energy. Here, it is shown that a GRES-scheduled sensor node will stay operational even in scenarios of mispredicted (i.e., overestimated) harvested energy.

II. Related Work

The authors of [5], [7] proposed an Integer Linear Programming approach to determine a schedule of configurations of different quality for solar energy harvesting sensor nodes over a day. However, this method assumes an accurate profile of the harvested energy for the considered time window. The work [4] solves a constrained optimization problem on traces of energy harvested produced by a weather-aware prediction model leveraging CNNs.

Other proposed energy management strategies for sensor nodes do not require the prediction of harvested energy, as they dynamically react to the variation in harvested energy. Some of these prediction-free energy management strategies are trained through Reinforcement Learning [2], [8]–[10]. Although these approaches have shown to obtain quite good results, the fact that they have to be trained on a Energy Harvesting (EH) dataset can make their deployment to various locations costly. Additionally, the neural network inference computations involved in online decision-making may increase the energy requirements. While [8] computes an allowed amount of energy that can be used in each time step based on the history of harvested energy, the authors leave out the way this energy budget can be used to maximize QoS. [2] suggests to adapt the power consumption to the available energy by adjusting the transmission rate (i.e., the frequency at which new data samples are transmitted) to a sink, trying to maximize the QoS while satisfying energy constraints.

Some work [11]–[15] base their decision-making not only on the harvested energy but also on a utility function, which associates a real value to a collected sample, e.g., the time elapsed since the collection of the sample. [13] assumes that a monitored value that has a high variability needs a high duty cycle to achieve maximal accuracy. [14] introduces an application-specific methodology that tries to minimize the waiting time at a parking lot by defining utility as its occupancy. However, these approaches are either very application- or platform-specific and need a utility to be defined, which requires domain knowledge and assumptions about the data.

Table I presents a qualitative comparison of related work.

TABLE I: Qualitative Comparison of Related Work

Work	Horizon	Predicted EH	Control Knob
[1]	Predictive	✓	Duty cycle
[5]	Predictive	✓	QoS
[2]	Reactive	✓	Transmission rate
[16]	Reactive	✓	Transmission rate
[17]	Reactive	✗	Duty cycle
[13]	Reactive	✗	Duty cycle
[4]	Reactive	✓	Energy allocation

III. Methodology

In the following, we introduce our models and assumptions about the sensor node and its workload and then present our novel scheduling algorithm GRES.

A. Model

We assume given a sensor node with a fixed (set) of periodically executed tasks (e.g., data logging, neural network inference, data transmission) with a period of T s. Each task can be executed at different levels of quality, called *configurations*, e.g., data rate, sampling frequency, amount of data sampled within time interval T, and bit width of each data sample. We assume given (e.g., by measurement) that a sensor node executing a task in configuration $c_i \in C$ consumes an amount $E_{\text{cons}}(c_i)$ of energy and yields a *quality* $q : C \to \mathbb{R}$.

We consider the sensor node to be powered by a battery of capacity E_{max}, which is refilled by energy harvested from an unreliable source (e.g., kinetic energy, solar energy). Let $E_{\text{harv}}(n)$ be the energy harvested during the n-th period. The state of charge of the battery $E_{\text{rem}}(n+1)$ at time $(n+1)\cdot T$ can then be computed as

$$E_{\text{rem}}(n+1) = min\{E_{\text{max}}, [E_{\text{rem}}(n) + E_{\text{harv}}(n) - E_{\text{cons}}(c_n)]^+\}$$

where $[x]^+ = max\{x, 0\}$.

For defining safety properties of operation, we first need to define a *time horizon* N as the number of periods in which a profile of harvested energy repeats. For example, for solar-harvested energy, the time horizon is typically $N \cdot T = 24$ h. Next, we define a *schedule* given a time horizon N as a sequence $c_n,\ n \in \{0, 1, \ldots, N-1\}$ of configurations. Let E_{min} denote the minimal energy level of a battery to operate correctly without risking a power failure, then a schedule is considered as *safe* for a time horizon N iff $E_{\text{rem}}(n) \geq E_{\text{min}} \forall n \in [0, N-1]$.

In this context, [1] shows that an effective way for a sensor node to operate indefinitely is to operate in an energy-neutral way given a time horizon N, i.e., the energy available at the end of the window $E_{\text{rem}}(N)$ has to be greater or equal than the energy at the start of the window $E_{\text{rem}}(0)$ to reach Energy Neutral Operation (ENO). With this modeling, we can then define the following optimization problem:

$$\max \quad Q = \sum_{n=0}^{N-1} q(c_n) \tag{1a}$$

$$\begin{aligned} \text{s.t.} \quad E_{\text{rem}}(n+1) = min\{&E_{\text{max}}, [E_{\text{rem}}(n) + E_{\text{harv}}(n) \\ &- E_{\text{cons}}(c_n)]^+\} \quad \forall n \in [0, N-1] \end{aligned}$$

$$E_{\text{rem}}(n) \geq E_{\text{min}} \quad \forall n \in [0, N-1] \tag{1b}$$

$$E_{\text{rem}}(N) \geq E_{\text{req}} \tag{1c}$$

The ILP with variables c_n maximizes the average quality of a schedule (Eq. (1a)) such that the remaining energy $E_{\text{rem}}(n)$ is never lower than E_{min}. It generalizes a model proposed in [1] in terms Eq. (1b) and Eq. (1c) in not requiring that $E_{\text{rem}}(N) = E_{\text{rem}}(0)$ (ENO), but defining a variable parameter $E_{\text{min}} \leq E_{\text{req}} \leq E_{\text{max}}$.

But not only is it computationally expensive to solve such an ILP in situ on a sensor node even if calculated only once per time horizon (e.g., a day). As the ILP is using a sequence of predicted harvested energy values $E_{\text{harv}}(n)$ which might be incorrect, the resulting schedule might be violating even the constraints in Eqs. (1b) and (1c), rendering it unsafe.

In order to deal with the variations and uncertainty in predicted energy traces we therefore introduce Guaranteed Remaining Energy Scheduling (GRES), a novel algorithm for dynamically controlling the configuration of a sensor node at runtime, using only the current value of E_{rem} at the beginning of the nth period of a time horizon consisting of N periods.

B. Guaranteed Remaining Energy Scheduling (GRES)

The strategy and safety guarantee implied by GRES can be simply described as follows: Starting at period $n = 0$ within each time horizon of N periods, calculate at the beginning of period n the highest quality setting c_n that, when kept constant for the remaining $N - n$ periods, would not violate the generalized ENO requirement of having still at least E_{rem} energy left at the end of the time horizon, thus being safe by design. GRES thereby simply computes:

$$c_n \leftarrow \text{argmax}\{q(c) | E_{\text{rem}}(n) - E_{\text{cons}}(c) \cdot (N - n) \geq E_{\text{req}}\} \quad (2)$$

In other words, GRES determines and assigns at the beginning of each period n of a time horizon of N periods the highest configuration c_n that, even in the worst case of not harvesting any additional energy until the end of the time horizon, would lead to a remaining amount of energy of at least E_{req} while steadily operating at this constant quality. Three examples of daily schedules of configurations generated by GRES and the ILP model of [1] for different traces of energy harvested and different starting energy levels $E_{\text{rem}}(0)$ are shown as examples in Figure 1.

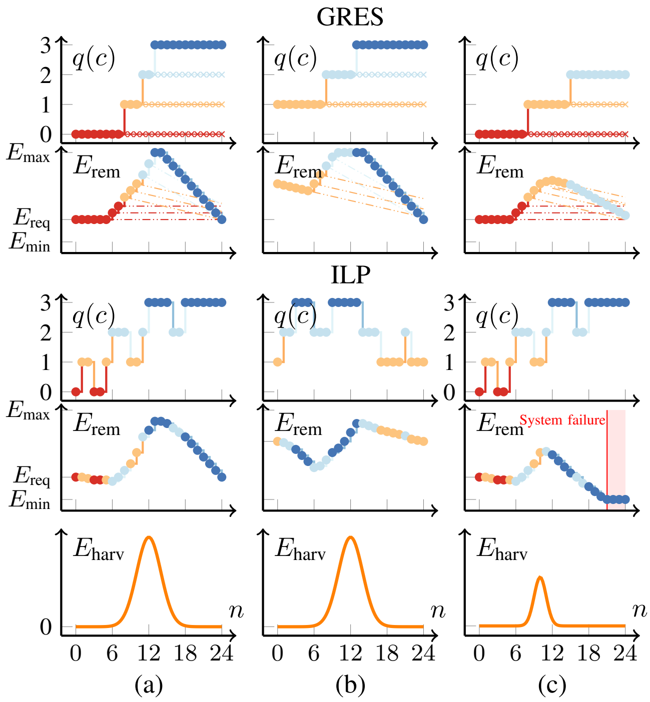


Fig. 1: Operation and comparison of GRES and ILP-generated schedules according to Eq. (1) for 3 scenarios of harvested energy and initial levels of available energy for a time horizon of $N = 24$ h. For scenario (a), where little energy is left initially at the start of the day $(n = 0)$, GRES yields an overall quality $Q = 43$, while the ILP's quality $Q = 50$. But already in (b), where the battery level at the start of the day $E_{\text{rem}}(0)$ is high, the requirements set by the ILP (in Eq. (1c)) leads to $Q = 42$ whereas for the same scenario, GRES achieves $Q = 50$. Finally, scenario (c) shows what happens if the predicted harvesting profile as used by the ILP deviates from the real scenario. Particularly if the values of E_{harv} are lower than expected, the ILP-based configuration schedule leads to a system failure while GRES operates safely throughout the day.

IV. Experiments

A. Generic representative case study

According to Eq. (1b), the safety guarantee of GRES as introduced before holds only if the consumed energy $E_{\text{cons}}(c)$ can be computed accurately in each configuration c. In the following, we present a quite general scenario of a sensor node application: [1] within a period of length T, let the task of the sensor node be to first sample a sensor signal of duration d_s with a sampling period T_s and a bit width b . The sensor node wirelessly transmits the sampled trace to a sink/gateway for d_p seconds through an interface of bandwidth μ and stays in deep sleep for the remaining time d_r such that $d_p + d_s + d_r = T$. Throughout its execution, a sensor node will continuously harvest energy from a photovoltaic module of peak power ϕ.

In the following, each configuration $c \in C$ is characterized by a tuple $(T_s(c), b)$. Moreover, we define the quality $q(c)$ of a configuration c as the bitrate $q(c) = b(c)/T_s(c)$. This is illustrated in Figure 2.

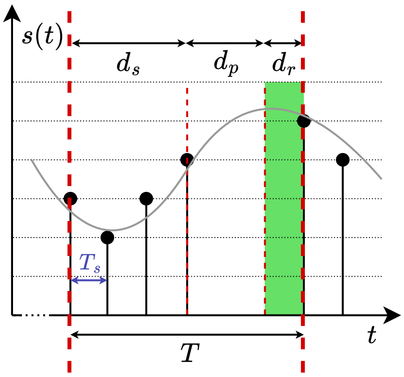


Fig. 2: Workload and execution model of a task of a sensor node. In each period T, data is sampled every T_s time units. After sampling (interval d_s), the data is processed (e.g., filtered, compressed, and stored or uploaded) during a time interval d_p. For the remaining time d_r, the power manager can set the sensor node to sleep mode to save energy.

Finally, we can compute the energy $E_{\text{cons}}(c)$ consumed by the sensor in a period when in configuration c as

$$E_{\text{cons}}(c) = d_s \cdot P_s + d_p(c) \cdot P_p + P_r \cdot (T - d_s - d_p(c)) \quad (3)$$

with P_s, P_p, P_r being the power consumptions of the node while sampling, transmitting and sleeping, respectively. In Eq. (3), $d_p(c) = (d_s \cdot q(c))/\mu$ denotes the cumulative time spent transmitting the sampled trace assuming a transmission bandwidth μ.

B. Experimental Setup

Based on the above fundamentals and application models, we can now evaluate the algorithms of GRES and the ILP more closely for realistic energy harvesting traces and considering

[1]Of course, other application scenarios can also be modeled and the safety guarantee of GRES holds, if only under such other scenarios of application and operation $E_{\text{cons}}(c)$ can be accurately computed or at least safe upper bounds for each configuration c.

a time horizon of one day (24 h) obtained from PVGIS[2] at the GPS coordinates 57°32’45.5”N 15°11’48.9”E. Here, we chose daily profiles for 3 days of winter ($22^{\text{nd}}, 23^{\text{rd}}$ and 24^{th} of February). For GRES as well as the ILP model given in Eq. (1) with Eq. (1c) replaced by the original constraint $E_{\text{rem}}(N) \geq E_{\text{rem}}(0)$ according to [1], we evaluate for each day, the average quality $Q_{\text{avg}} = Q/N$. We compute these values for multiple values of $E_{\text{rem}}(0)$ and E_{req}. Table II shows the parameters used during these experiments along with the available configurations c and characterized consumed energy $E_{\text{cons}}(c)$ and bit rate $q(c)$.

TABLE II: Parameters and Configurations Used During the Experiments

(a) Parameters

Parameter	Value
d_s [s]	630
T [s]	900
P_s [mW]	10
P_p [mW]	40
P_r [mW]	5
μ [kbps]	100
$E_{\max}$ [J]	93600
ϕ [W]	1.9

(b) Configurations

b_d	c T_s [µs]	E_{cons} [J]	q [kbps]
0	0	0.00	0
8	40	51.75	200
9	40	57.26	225
8	33	60.57	242
10	40	62.77	250
9	33	67.18	272
8	28	69.39	285
10	33	73.80	303
9	28	77.10	321
10	28	84.82	357

C. Analysis of Different Start ($E_{rem}(0)$) and Minimal End Energy Requirements (E_{req})

We now compare the quality achieved by GRES for three winter days with those of ILP-based schedules (see Section III-A) computed based on a daily energy harvesting profile determined as the average daily profile of days belonging to the same winter month over 4 consecutive years. The results are presented in Figure 3 showing that while the solutions generated by the ILP at times do provide an overall higher quality than GRES, there are also many settings of $E_{\text{rem}}(0)$ and E_{req} for which the ILP schedule applied to the actual harvested energy data, leads to system shutdowns and thus interruptions of service (marked with ×). Note that this does not happen for any setting when using GRES.

GRES provides continued operation, if possible, by choosing at each time the configuration that has the highest quality without violating the constraint of a minimal remaining energy E_{rem} and the end of the time horizon. For example, even at low quality, the IoT system will still deliver data continuously without any loss of service.

Finally, Figure 4 shows a scenario of 4 consecutive days in winter during which the observed harvested energy profiles per day were lower than forecasted. Shown are the values of E_{rem}, E_{harv} and $q(c)$ obtained for GRES and the ILP-based schedules computed for each of four consecutive winter days based on forecasted energy harvesting profiles (dashed blue curves in Figure 4) for the setting $E_{\text{rem}}(0) = E_{\text{req}} = 0.4 \cdot E_{\max}$.

[2] https://re.jrc.ec.europa.eu/pvg_tools/en/

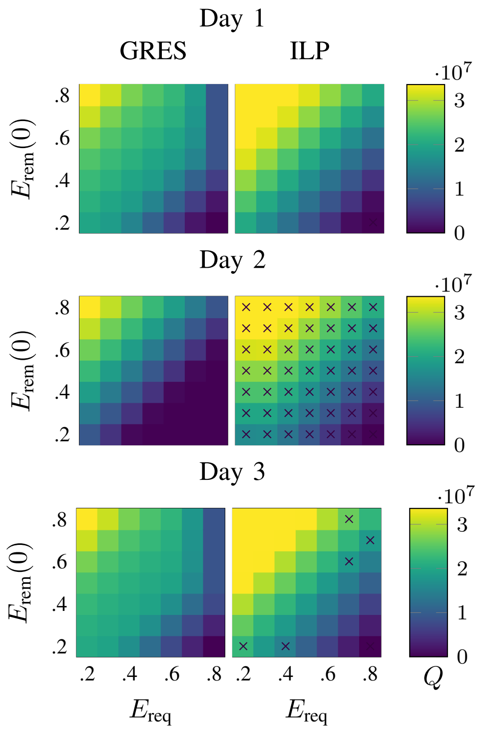


Fig. 3: Values of Q obtained per day by GRES and by the ILP for 3 days of winter, in dependence of different initial conditions $E_{\text{rem}}(0)$ and minimal end-of-day energy requirements E_{req}. The × marks show when the computed and applied configuration schedule leads to a low battery state and thus system shutdown and interrupt of service.

It can be noticed that if the amount of harvested energy is lower than assumed during planning, the ILP-based scheduler applies configurations that are too energy-consuming which may lead the sensor node to fully consume its battery energy after the second day (Figure 4 middle) leading to a shutdown.

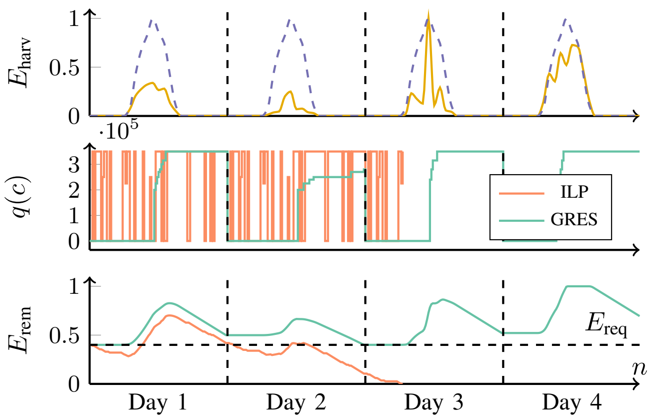
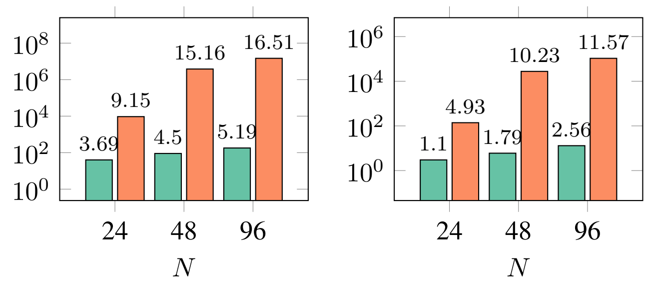


Fig. 4: Comparison of GRES and ILP-generated schedules over 4 consecutive days of winter. Where the ILP-generated schedule assumes a given 24 h forecast (dashed blue) and fails when subject to a real (observed) harvesting profile (brown), GRES dynamically reacts to the current profile.

D. Overhead Analysis

Finally, we analyze and compare the overheads of computing schedules for both GRES and efficient implementation of the ILP optimization problem in Eq. (1) using dynamic programming in [7] for different values of N in terms of execution time and energy requirements. The measurements of execution time and consumed energy were taken for implementations of the two methods on a Texas Instruments MSP430FR2433 platform using the open-source integrated development environment (IDE) Energia [18]. The cumulative execution time of GRES for a time horizon of N invocations time horizon is compared to one execution of the algorithm in [7] that returns a schedule of N configurations. Similarly, the cumulative energy consumption is shown in Figure 5.



(a) Execution time [µs] (b) Energy consumed [µJ]

Fig. 5: Comparison of execution time and energy consumption overheads of GRES (in green) and the algorithm in [7] (in orange) on an MSP430 platform for different values of N.

As can be seen, even the more efficient solution of the ILP model in Eq. (1) using dynamic programming [7] requires two to four orders of magnitude more energy than GRES.

V. Conclusion

The increasing workload in terms of compute-intensive tasks on energy-harvesting wireless sensor nodes brings the need for new energy management strategies. In this paper, we propose a quite unique, but still simple energy management strategy that dynamically determines and assigns a maximal quality configuration that can be sustained until the end of the time horizon while guaranteeing that at the end of the time horizon, a given minimal amount of energy is still remaining. To the best of our knowledge, this is the first time that such guarantees can be given. It has been shown in experiments that previous non-adaptive approaches relying on forecasts of harvested energy, i.e., an ILP-based scheduler, not only suffer from long computation times, but can lead to unexpected system shutdowns if the prediction of the harvested energy profiles is incorrect, thus loss of service.

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